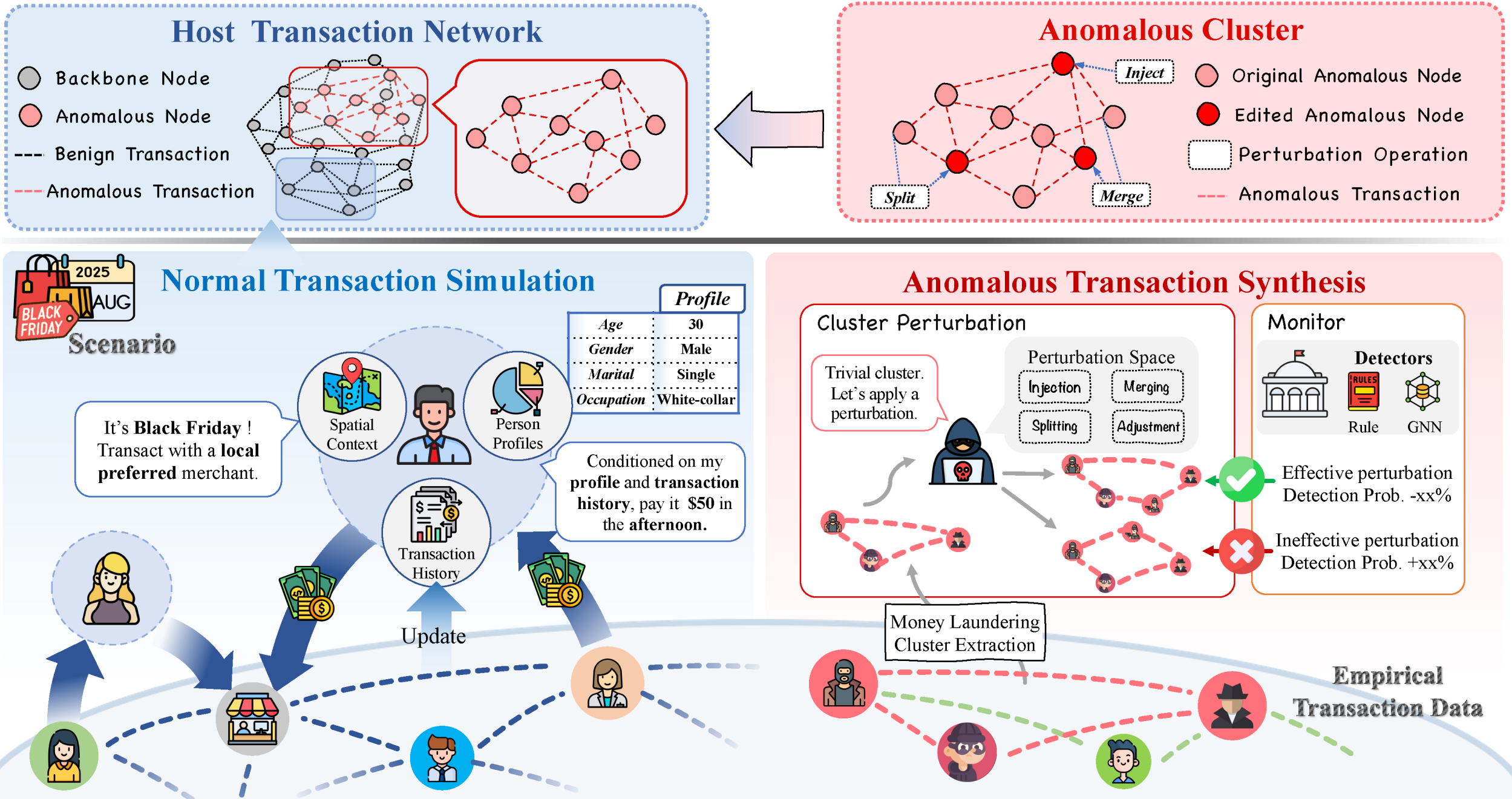

Host Transaction Network
Backbone Node
Anomalous Node
Benign Transaction
Anomalous Transaction
Anomalous Cluster
Inject
Split
Merge
Original Anomalous Node
Edited Anomalous Node
Perturbation Operation
Anomalous Transaction
Normal Transaction Simulation
2025
AUG
BLACK FRIDAY
Scenario
Profile
Age 30
Gender Male
Marital Single
Occupation White-collar
Spatial Context
Person Profiles
Transaction History
It's Black Friday ! Transact with a local preferred merchant.
Conditioned on my profile and transaction history, pay it $50 in the afternoon.
Update
Anomalous Transaction Synthesis
Cluster Perturbation
Trivial cluster. Let's apply a perturbation.
Perturbation Space
Injection
Merging
Splitting
Adjustment
Monitor
Detectors
Rule
GNN
Effective perturbation
Detection Prob. -xx%
Ineffective perturbation
Detection Prob. +xx%
Money Laundering Cluster Extraction
Empirical Transaction Data